



Module 2

Chapter 4



3 x 3

Rank, Transpose, Determinant and Inverse

2 → 1 in 3 zero

1 → zeroed out

Rank

The *Rank* of a Matrix is a fundamental concept in linear algebra that measures the **number of linearly independent rows or columns in a matrix**. It essentially determines the **dimensionality of the vector space formed by the rows or columns of the matrix**. It also helps us determine if a system of equations has a unique solution.

The rank of a matrix is denoted using $\rho(A)$ where A is any matrix. ***The rank of the matrix cannot exceed the total number of rows in a matrix.*** So, if we have a 4×4 matrix, the Rank of the matrix cannot be more than 4.

How to calculate the Rank of a Matrix?

One way to find the rank of a matrix is to perform *Gauss-Jordan elimination* on it.

Recall that Gauss-Jordan elimination will **zero out** rows that are **linearly**

dependent on other rows, leaving only rows that are linearly independent

from each other. So, by the time the halfway point is reached in Gauss-Jordan

elimination, where we've zeroed out all the rows that will be zeroed out, we can tell what the rank of the matrix is. ***The Rank of the matrix becomes the number of non-zero rows remaining.***

Let's look at an example

Let's try to find the rank of this matrix:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Applying Gauss Jordan Elimination will give us:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

There's only one nonzero row, so the rank is 1. And indeed, there's only one

linearly independent vector here, $[1 \ 1 \ 1]$. we can *zero out a row* in Gauss-Jordan elimination if and only if it is *linearly dependent* upon the other rows that haven't been zeroed. The above could be defined as Row Rank. But it is also possible that the columns of a matrix are linearly dependent. The number of linearly independent columns of a matrix gives us the column Rank. For any given matrix, the **row rank is always equal to the column rank**. This is a useful property especially when a column of a matrix is visibly dependent on the other columns while the row dependence is not too obvious. We can easily determine if the matrix is singular without doing Gauss-Jordan Elimination.

Let's look at an example:

$$\begin{bmatrix} 1 & 4 & 5 \\ 7 & 1 & 8 \\ 11 & 2 & 13 \end{bmatrix} \quad \begin{array}{l} \leq 2 \\ \leq 2 \\ \# \text{ columns} \rightarrow 3 \end{array}$$

This matrix isn't obviously of reduced rank if we look at the rows, but if we look at the columns, we can see that the third column is a sum of the previous two, and so this matrix must be singular.

If the rank of a matrix is equal to the number of unknowns (columns), then the matrix is nonsingular, and if a solution exists, it will be unique.

Every missing unit of rank corresponds to a linearly dependent row of the matrix, which in turn means that the solution has one more degree of freedom to play with.

Here are some equivalent statements about an $n \times n$ matrix A with rank n :

- A is nonsingular.
- The rows of A are linearly independent.
- The columns of A are linearly independent.
- Any linear system with matrix of coefficients A has a unique solution.

Let's Practice

Find the rank of the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\begin{aligned} R_2 &= R_2 - 4R_1 \\ R_3 &= R_3 - 7R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$$

$$R_3 = R_3 - 2R_2$$

Row $\rightarrow$ Echelon form

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 = R_2 / -2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_1 = R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \rho(A) = 2$$

- * Echelon form:
- 1) Every Row needs a Pivot
 - 2) Pivot need not be 1 (but has to be non-zero)
 - 3) Pivot of a Row should be to the right of the pivot of the Row above it

Transpose of a matrix

The transpose of a matrix A is represented by A^T and is obtained by swapping the rows and columns of the matrix.

So, if we have a matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

Then $A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

The dimensions/order of the transpose of an $m \times n$ matrix is $n \times m$

Some important properties of the transpose of a matrix:

- A square matrix “ A ” of order “ $n \times n$ ” is said to be **orthogonal matrix**, if $AA^T = A^T A = I$, where “ I ” is an identity matrix of order “ $n \times n$.”
- A square matrix “ A ” of order “ $n \times n$ ” is said to be a **symmetric matrix** if its transpose is the same as the original matrix, i.e., $A^T = A$.
- A square matrix “ A ” of order “ $n \times n$ ” is said to be a **skew-symmetric matrix** if its transpose is equal to the negative of the original matrix, i.e., $A^T = -A$.
- **Double Transpose of a Matrix:** Transpose of the transpose matrix is the original matrix itself.

$$(A^T)^T = A$$

- **Transpose of product of Matrices:** This property says:

$$(AB)^T = B^T A^T \quad *$$

- **Transpose of addition of Matrices:** This property says:

$$(A + B)^T = A^T + B^T$$

Let's Practice

Determine if the following matrices are symmetric or skew symmetric:

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 5 \end{bmatrix} \text{ symmetric}$$

$$\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix} \text{ skew symmetric}$$

Determinant of a Matrix

The **determinant of a matrix** is a *scalar value* that can be calculated for a **square matrix** and gives us information about the singularity of matrix and is **zero iff A is singular** (and not invertible). When a determinant is not zero, it reflects how much the matrix scales its inputs; but that's not usually as useful as the fact that its being zero indicates that a matrix is singular. It's often more computationally efficient to find a determinant than to use other methods of determining singularity, such as finding the rank with Gauss-Jordan.

The notation for determinant of a matrix A is :

Det A or |A|. It is also denoted by the symbol Δ

What can you say about the rank of a square matrix with a non-zero determinant?

Determinant of a 2 * 2 Matrix:

Let A be a 2*2 matrix: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $|A| = ad - cb$

For larger matrices we use the concept of *co-factor expansion*, among other methods. We need to know two concepts to understand co-factor expansion:

Minor of element of matrix and co-factors:

Minor of element of Matrix:

Minors are the determinants for every element of a matrix obtained by eliminating the rows and columns of that element. So, if we have a matrix:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$C_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

Then the minor of a_{12} will be given by M_{12} :

$$\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

Co-factor of element of a matrix:

For an element a_{ij} , the cofactor of this element is C_{ij} and the minor is M_{ij} then, the cofactor can be written as:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Let's look at an example:

What is the minor and cofactor of the element placed in the first row and second column of the determinant:

$$\begin{vmatrix} 2 & 1 & 2 \\ 4 & 5 & 0 \\ 2 & 0 & 1 \end{vmatrix}$$

First row second column element is: 1

$$\text{So, the minor will be } M_{12} = \begin{vmatrix} 4 & 0 \\ 2 & 1 \end{vmatrix} = 4*1 - 2*0 = 4$$

$$\text{Now, Co-factor will be } C_{12} = (-1)^{1+2} M_{12} = (-1)*4 = -4$$

The determinant of a Matrix is calculated along any one row or column of a Matrix by multiplying the elements of the row/column to their co-factors and adding them.

It is always easier to choose the row/column with the maximum number of zeroes to reduce the amount of computation.

Let's look at an example:

$$\text{Find det } |A| = \begin{vmatrix} 1 & 3 & 0 \\ 4 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix}$$

We see the third column has the most number of zeroes, so we will expand along this column:

$$|A| = a_{13} * C_{13} + a_{23} * C_{23} + a_{33} * C_{33}$$

$$a_{13} * C_{13} = 0 * (-1)^{1+3} \begin{vmatrix} 4 & 1 \\ 2 & 0 \end{vmatrix} = 0$$

$$a_{23} * C_{23} = 0 * (-1)^{2+3} \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = 0$$

$$a_{33} * C_{33} = 1 * (-1)^{3+3} \begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} = -11$$

$$\text{Thus, } |A| = 0 + 0 - 11 = -11$$

Determining the determinant of a higher order matrix involves more complex methods such as expansion by minors which require breaking down the matrix into smaller submatrices and recursively finding their determinants.

Let's look at the determinant of a 4*4 matrix:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$\text{Det } |A| = (-1)^{1+1}a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} + (-1)^{1+2}a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} +$$

$$(-1)^{1+3}a_{13} \begin{vmatrix} a_{22} & a_{22} & a_{24} \\ a_{32} & a_{32} & a_{34} \\ a_{42} & a_{42} & a_{44} \end{vmatrix} + (-1)^{1+4}a_{14} \begin{vmatrix} a_{22} & a_{23} & a_{23} \\ a_{32} & a_{33} & a_{33} \\ a_{42} & a_{43} & a_{43} \end{vmatrix}$$

Let's Practice

Find the determinant of the following Matrices:

$$\begin{vmatrix} 3 & 3 & 0 \\ 2 & 1 & 1 \\ 5 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} \overset{+}{0} & \overset{-}{0} & \overset{+}{3} & \overset{-}{3} \\ \overset{-}{3} & \overset{+}{0} & \overset{-}{1} & \overset{+}{2} \\ \overset{+}{1} & \overset{-}{0} & \overset{+}{2} & \overset{-}{4} \\ \overset{-}{2} & \overset{+}{1} & \overset{-}{3} & \overset{+}{2} \end{vmatrix}$$

Inverse of a Matrix

The **inverse of a Matrix “A”**, denoted as A^{-1} , is **another matrix that, when multiplied by A**, results in the **identity matrix I**. In mathematical terms:

$$A \times A^{-1} = A^{-1} \times A = I$$

$$\begin{vmatrix} 3 & 3 & 0 \\ 2 & 1 & 1 \\ 5 & 0 & 1 \end{vmatrix}$$

selecting Row 3:

$$5 \times (-1)^{3+1} (3-0) + 0 + 1 \times (-1)^{3+3} (3-6)$$

$$= 5 \times 3 - 1 \times 3$$

$$= 15 - 3 = 12$$

$$\begin{vmatrix} 0 & 0 & 3 & 3 \\ 3 & 0 & 1 & 2 \\ 1 & 0 & 2 & 4 \\ 2 & 1 & 3 & 2 \end{vmatrix}$$

selecting Row 1

$$3 \times (-1)^{1+3} \begin{vmatrix} 3 & 0 & 2 \\ 1 & 0 & 4 \\ 2 & 1 & 2 \end{vmatrix} + 3 \times (-1)^{1+4} \begin{vmatrix} 3 & 0 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 3 \end{vmatrix}$$

$$3 \left\{ 3 \times (-1)^{1+1} (-4) + 2 \times (-1)^{1+3} (1) \right\} - 3 \left\{ 1 \times (-1)^{2+1} (-1) + \right.$$

$$\left. 2 \times (-1)^{2+3} (3) \right\}$$

$$= 3 \left\{ -12 + 2 \right\} - 3 \left\{ 1 - 6 \right\}$$

$$= 3 \times -10 + 3 \times 5$$

$$= -30 + 15 = -15$$

Note: Not all matrices have an inverse. A matrix must be **square** (same number of rows and columns) and must be **non-singular** (its determinant is not zero) to have an inverse.

Finding Matrix Inverses:

We will learn two ways of computing Matrix Inverses:

- Determinant Method
- Elementary Transformation method

Determinant Method

For determinant method, we need to know about the *Adjoint of a matrix*.

Adjoint of a square matrix $A = [a_{ij}]_{n \times n}$ is defined as the **transpose of the matrix** $[A_{ij}]_{n \times n}$, where A_{ij} is the cofactor of the element a_{ij} . In other words, the transpose of a cofactor matrix of the square matrix is called the adjoint of the matrix. The adjoint of the matrix A is denoted by **adj A**. This is also known as *adjugate matrix* or *adjunct matrix*.

So, if we have a 2*2 Matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^T$$

Then

$$\text{Adj } A = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix}$$

Where

A_{11} = Cofactor of a_{11}

A_{12} = Cofactor of a_{12}

A_{21} = Cofactor of a_{21}

A_{22} = Cofactor of a_{22}

Similarly, if we have a 3*3 Matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then,

$$\text{Adj } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

Let's take an example:

Let's try to find the adjoint of the matrix $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

So the cofactors are:

$$A_{11} = a_{22} = 4 \quad ((-1)^{1+1} \times 4)$$

$$A_{21} = -a_{12} = -3 \quad ((-1)^{2+1} \times 3)$$

$$A_{12} = -a_{21} = -1 \quad ((-1)^{1+2} \times 1)$$

$$A_{22} = a_{11} = 2 \quad ((-1)^{2+2} \times 2)$$

So, the adjoint of the matrix will be $\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$

Let's Practice:

Find the adjoint of the given Matrix:

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 3 \end{bmatrix}$$

Now that we know how to calculate the adjoint of a matrix, we can go ahead and

define the formula for finding the *inverse of a matrix*:

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 3 \end{bmatrix}$$

Calculating Co-factors:

Row 1

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 5 \\ 0 & 3 \end{vmatrix} = 9$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} = -1$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 1 & 0 \end{vmatrix} = -3$$

Row 2

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -1 & 2 \\ 0 & 3 \end{vmatrix} = 3$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = -1$$

Row 3

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 2 \\ 3 & 5 \end{vmatrix} = -11$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = -1$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 5$$

∴ Co-factor Matrix

$$\begin{bmatrix} 9 & -1 & -3 \\ 3 & 1 & -1 \\ -11 & -1 & 5 \end{bmatrix}$$

∴ adj A =

$$\begin{bmatrix} 9 & 3 & -11 \\ -1 & 1 & -1 \\ -3 & -1 & 5 \end{bmatrix}$$

$$\mathbf{A}^{-1} = \frac{\text{Adj}(\mathbf{A})}{\text{Det}(\mathbf{A})}$$

So, for a 2 * 2 matrix: $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Similarly, for a 3*3 matrix: $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$

Let's Practice:

Find the ~~determinant~~ of the following matrices using determinant method:

Inverse

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 2 & 3 \\ 0 & 0 & 4 \\ 2 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{bmatrix}$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{vmatrix} = -36 \quad (\text{using co-factor expansion})$$

$$\text{Cofactor Matrix} = \begin{bmatrix} 33 & 6 & -27 \\ -12 & -12 & 12 \\ -3 & 6 & -3 \end{bmatrix}$$

$$\therefore \text{adj Matrix} = \begin{bmatrix} 33 & -12 & -3 \\ 6 & -12 & 6 \\ -27 & 12 & -3 \end{bmatrix}$$

$$\therefore \text{Inverse:} \begin{bmatrix} 33/-36 & -12/-36 & -3/-36 \\ 6/-36 & -12/-36 & 6/-36 \\ -27/-36 & 12/-36 & -3/-36 \end{bmatrix}$$

$$= \begin{bmatrix} -11/12 & 1/3 & 1/12 \\ -1/6 & 1/3 & -1/6 \\ 3/4 & -1/3 & 1/12 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 2 & 3 \\ 0 & 0 & 4 \\ 2 & 0 & 0 \end{bmatrix}$$

Determinant of Matrix = 16

$$\text{adj Matrix} = \begin{bmatrix} 0 & 0 & 8 \\ 8 & -6 & -24 \\ 0 & 4 & 0 \end{bmatrix}$$

$$\therefore \text{Inverse Matrix} = \begin{bmatrix} 0 & 0 & 1/2 \\ 1/2 & -3/8 & -3/2 \\ 0 & 1/4 & 0 \end{bmatrix}$$

Elementary Transformation Method

We can also find the inverse of a matrix using elementary Row operations that we learnt for Gauss Jordan Elimination. Here are the steps for getting the inverse of a given Matrix:

Step 1: Write the given matrix as $A = IA$, where I is the *identity matrix* of the order/dimensions same as A .

$$AA^{-1} = I$$
$$IA^{-1}A = I$$

Step 2: Use the sequence of *valid row operations* till we get an identity matrix on the LHS (keep applying the same elementary operations on the identity matrix in RHS). We will end up getting the following equation: $I = BA$. We know that $AA^{-1} = I$. So, *matrix B on RHS is the inverse of matrix A .*

$$\begin{array}{c} \textcircled{A} = \textcircled{I} A \\ \downarrow \\ I = \textcircled{B} A \end{array}$$

Let's take an example:

Let us try to find the inverse of the following matrix: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$$I = \underline{B} A$$

Step 1: Set up $A = IA$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 2: Apply elementary operations till we get I on the LHS:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$R_1 = R_1/2$$

$$\begin{bmatrix} 1 & 1/2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \times A$$

$$R_2 = R_2 - R_1$$

$$\begin{bmatrix} 1 & 1/2 \\ 0 & 3/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix} \times A$$

$$R_2 = 2/3 R_2$$

$$\begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ -1/3 & 2/3 \end{bmatrix} \times A$$

$$R_1 = R_1 - R_2/2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix} \times A$$

$$\text{Thus } A^{-1} = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

Let's practice:

Find the inverse of the matrices using elementary operations:

a. $\begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 5 & 1 \end{bmatrix}$

$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 7 & 4 & 0 & 1 \end{array} \right]$

$$A A^{-1} = I$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} \\ a_{21}b_{11} + a_{22}b_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \dots (i)$$

$$\begin{bmatrix} a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \dots (ii)$$

From (i): we can solve for b_{11} and b_{21}

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Augmented Matrix:

$$\left[\begin{array}{cc|c} a_{11} & a_{12} & 1 \\ a_{21} & a_{22} & 0 \end{array} \right]$$

From (i) we solve for b_{12} and b_{22}

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Augmented Matrix

$$\left[\begin{array}{cc|c} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 1 \end{array} \right]$$

Combining the 2 augmented matrices

$$\left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right]$$

we can then apply elementary Row operations

Q. $\begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times A$$

$$R_1 = R_1/2$$

$$\begin{bmatrix} 1 & 1/2 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \times A$$

$$R_2 = R_2 - 7R_1$$

$$\begin{bmatrix} 1 & 1/2 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ -7/2 & 1 \end{bmatrix} \times A$$

$$R_2 = 2R_2$$

$$\begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ -7 & 2 \end{bmatrix} \times A$$

$$R_1 = R_1 - R_2/2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ -7 & 2 \end{bmatrix} \times A$$

$$\therefore A^{-1} = \begin{bmatrix} 4 & -1 \\ -7 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 5 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times A$$

$$R_2 = R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 0 \\ 3 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times A$$

$$R_3 = R_3 - 3R_1, \quad R_2 = -R_2$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 2 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \times A$$

$$R_1 = R_1 - R_2, \quad R_3 = R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix} \times A$$

$$R_3 = R_3 / -5, \quad R_1 = R_1 - 2R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -19/5 & 9/5 & 2/5 \\ 2 & -1 & 0 \\ 7/5 & -2/5 & -1/5 \end{bmatrix}$$

$\rightarrow A^{-1}$
 $\times A$